THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Sex of a child is determined by the father. Reason (R): Sperms carry either X or Y chromosome. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A pea plant RrYy is crossed with rryy. Out of 200 offspring, the number of round green plants is expected to be 1

(a) 100 **(b)** 25 **(c)** 150 **(d)** 50

3. A pea plant that is tall and round (TtRr) is crossed with dwarf wrinkled (ttrr). The offspring ratio is 1

(a) 9 : 3 : 3 : 1 **(b)** 1 : 1 **(c)** 1 : 1 : 1 : 1 **(d)** 3 : 1

4. Assertion (A): Acquired traits are not inherited by the next generation. Reason (R): Changes in body cells do not change the DNA of germ cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. In a cross Tt x Tt, three offspring are tall. The probability that the fourth is dwarf is 1

(a) 3/4 **(b)** 1 **(c)** 1/2 **(d)** 1/4

6. In pea plants F2 seeds show round-green and wrinkled-yellow types that were absent in both parents. This happens due to 1

(a) blending of traits
(b) loss of genes in F1
(c) acquired traits
(d) new combinations of independently inherited genes

7. Assertion (A): A recessive trait is not seen in the F1 generation. Reason (R): The dominant allele masks the recessive allele in heterozygotes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A tall plant is crossed with a dwarf plant; 52 of 100 offspring are tall. The tall parent was 1

(a) TT **(b)** Tt
(c) tt **(d)** cannot be determined

9. **Assertion (A):** When pure tall (TT) and pure dwarf (tt) pea plants are crossed, all the F1 plants are tall. 1
Reason (R): Tallness is the dominant trait over dwarfness.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Both parents contribute equally to the genetic material of a human child. 1
Reason (R): The sex of a human child is determined by the sperm that fertilises the egg.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In a cross of two heterozygous tall plants (Tt x Tt), 240 plants were produced. Find the number of tall plants that are homozygous. 2

12. Explain the reason a plant with 'Tt' genotype shows only the tall phenotype. 2

13. A man with attached earlobes (ff) and a woman with free (Ff) have 4 children. Expected number with attached earlobes? 2

14. A tall plant when crossed with a dwarf plant gives 30 tall and 28 dwarf plants. Give the genotypes of the parents. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A pea plant has yellow round seeds. How would you find its genotype? Describe the cross and the possible results. 3

16. How many chromosomes does each of the following have in humans: sperm, egg, zygote, liver cell? What is the number of autosomes in each? 3

17. Explain why every egg of a woman carries an X chromosome, though she has two X chromosomes. 3

OR

A couple has three sons. What is the probability that the fourth child is a girl? Explain how the answer is obtained. 3

18. In a population of beetles, a green colour variant arises through a DNA copying error. Birds eat red beetles more. Explain what happens to the proportion of green beetles over generations. 3

OR

Why is it possible for two brown-eyed parents to have a blue-eyed child, if blue is recessive? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita crossed a pure tall pea plant (TT) with a pure dwarf pea plant (tt). All the F1 plants were tall. She allowed the F1 plants to self-pollinate and obtained 480 F2 plants. 5
(i) Write the genotype of the F1 plants.
(ii) What is the F2 phenotypic ratio?
(iii) How many of the 480 F2 plants are expected to be dwarf?
(iv) How many of the 480 F2 plants are expected to be tall with genotype Tt? Give the genotypic ratio.

20. A tall plant with round seeds (TtRr) is crossed with a dwarf plant with round seeds (ttRr). Find the expected phenotypic ratio of the offspring. 5

OR

Discuss the sources of variation in a sexually reproducing population and show how variation can be useful for the species, by giving a human example. 5

END OF QUESTION PAPER — Total: 40 Marks